

EcoRAG Experiment C

Research Summary Report

EcoRAG EXPERIMENT C - RESEARCH SUMMARY Evidence Sufficiency & Citation Correctness Evaluation

GENERATED: 2026-10-04 DATASET: 27 Environmental & Sustainability Questions (Frozen) MODEL: Qwen2.5-3B-Instruct (CPU, bf16, greedy decoding) EMBEDDINGS: BGE bge-small-en-v1.5 (384 dimensions) REVIEW: Single-reviewer manual annotation DOCUMENTS: 9 PDFs, 1,242 chunks

EXECUTIVE SUMMARY

KEY FINDING: EcoRAG demonstrates consistent answer generation with strong evidence utilization (91% coverage) but MAJOR WEAKNESSES in: 1. CALIBRATION (70% accuracy) - fails to abstain when evidence is absent 2. CITATION CORRECTNESS (24% incorrect) - especially in multi-source questions

PERFORMANCE BY EVIDENCE CONDITION: • Sufficient evidence (14 questions): 100% correct calibration (14/14) ✓ • Insufficient evidence (9 questions): 44% abstention recall (4/9) ✗ • Overall: 70% calibration accuracy across 27 questions

CRITICAL RESULTS: ✓ Answer content fully correct: 18/27 = 67% ✗ Citations incorrect (manual review): 20/85 = 24% ✓ Evidence coverage in top-5: 32/35 = 91% ✗ Dependent-source citation errors: 7/9 = 78%

DETAILED METRICS (S1-S6)

S1: RETRIEVAL SUFFICIENCY (Evidence availability)
Sufficient (all required units in top-5): 14 questions = 52% Partial (some required units in top-5): 4 questions = 15% Insufficient (no required units): 9 questions = 33%
Unit Retrieval Recall: 35/45 = 78%

FINDING: About 1 in 5 required facts aren't retrieved. Manageable problem, but the larger issue is the system answering when evidence is absent.

S2: ANSWER-UNIT COVERAGE (Using retrieved evidence)
S2-gen (coverage of retrieved units): Pooled: 32/35 = 91% ✓ Per-question average: 0.89
S2-e2e (coverage of all required units): Pooled: 35/55 = 64% Per-question average: 0.62

FINDING: When evidence is available in top-5, the model uses 91% of it. When it's not available, it still tries to answer (S4 issue).

S3: CONCLUSION JUSTIFICATION (Citation quality)
S3a - Cited passages justify the claim: Justified: 10 claims Partial: 7 claims Not at all: 5 claims Mean score: 0.61 (out of 1.0)
S3b - Full top-5 justifies: Justified: 14 claims Partial: 3 claims Not at all: 5 claims Mean score: 0.70 (out of 1.0)

FINDING: 70% of answers are justified by their retrieved evidence. 30% lack proper justification (missing citations or wrong citations).

S4: CALIBRATED BEHAVIOUR (Knowing when to stop)
Expected / Actual: Gold Label → Model Behaviour
Sufficient evidence: Expected: FULL answer Actual: 14/14 FULL ✓ [100% correct] False abstentions: 0
Partial evidence: Expected: FULL answer + GAP statement Actual: 1/4 correct 2/4 answered fully anyway 1/4 abstained (wrong)
Insufficient evidence: Expected: ABSTAIN Actual: 4/9 ABSTAIN ✓ [44% correct] 5/9 answered fully anyway ✗ [56% over-answer]

OVERALL CALIBRATION ACCURACY: 19/27 = 70%
FINDING: Perfect when evidence exists (14/14). Fails catastrophically when evidence is weak (only 1/4 partial) or missing (only 4/9 abstain). Core problem: OVER-ANSWERING.

S5: OVER-REACH CLAIMS (Going beyond evidence)
Total over-reach claims: 9/86 = 10%
By type: O2 (entity/scope substitution): 5 claims O3 (unsupported generalization): 3 claims O4 (fabrication): 1 claim
Questions with ≥1 over-reach: 6/22 = 27%

EXAMPLES: • EMDE 2023 figure presented as Africa 2024 • German professional salary presented as Austrian • Projected warming presented as avoided warming

FINDING: About 1 in 4 answered questions contain at least one over-reach claim. Usually entity/scope substitution rather than pure fabrication.

S6: INDEPENDENT SUPPORT (Citation diversity)
Covered units with supporting cited chunks: 30/35 = 86% Mean distinct origins per covered unit: 0.86 Dependent multi-citations (same source 2+): 2 False corroborations: 0
FINDING: Most covered units have at least one supporting chunk. Low citation diversity (0.86) means claims rely heavily on single sources, not cross-corroborated.

CITATION CORRECTNESS (SEPARATE)

VERIFIER (Rule-based, deterministic): Claims with citations: 81/82 = 99% Misattributed (wrong source): 10/81 = 12% Potentially supported: 68/81 = 84%

MANUAL REVIEW (Human judgment, n=85 cited claims): Supported (including minor distortion): 65/85 = 76% Misattributed: 11/85 = 13% Unsupported: 9/85 = 11% INCORRECT (misattr + unsupported): 20/85 = 24% Per-question average: 7.22/22 = 0.33

DISAGREEMENT BETWEEN VERIFIER & MANUAL: 14 of 85 claims disagreed: • 4 were bare-value lines (A2 decision) • 10 were verifier misses (claimed supported but actually wrong)

EXAMPLE: Q18 cited USD 36,800 salary as Austrian, but it was for German professionals.

CONCENTRATION: Citation errors are concentrated in dependent-source questions. • Type (a) single-chunk: 0/13 = 0% error ✓ • Type (e) dependent: 7/9 = 78% error ✗

This suggests the system confuses which document supports which fact when multiple sources are involved.

FAILURE TAXONOMY

Analysis of 55 required answer units (priority order):

Retrieval Failure: 10/55 = 18% Unit in corpus but not in top-5. Retrievable but not retrieved.

Calibration Failure: 4/55 = 7% Evidence was weak; model over-answered anyway.

Generation Failure: 3/55 = 5% Unit was retrieved but not used in the answer.

Citation Failure: 5/55 = 9% Unit was used but cited incorrectly.

CORRECT: 33/55 = 60% Properly retrieved, covered, cited.

LARGEST ISSUE BY FAR: Retrieval (18%) + Calibration (7%) = 25% of failures SECONDARY: Citation errors (9%)

INSIGHT: The system's job is not primarily citation binding. The bigger problems are: (1) not retrieving everything, (2) answering when it shouldn't.

RESULTS BY EVIDENCE CONDITION

SUFFICIENT EVIDENCE (14 questions, all required units in top-5): Calibration accuracy (FULL): 14/14 = 100% ✓ Content fully correct: 13/14 = 93% All citations correct: 7/14 = 50% Over-reach: 4/57 = 7% Manual citation error rate: 14/56 = 25%

VERDICT: When evidence exists, the system performs excellently. 50% have all citations correct (surprisingly low but acceptable).

PARTIAL EVIDENCE (4 questions, some required units in top-5): Calibration accuracy (PARTIAL+GAP): 1/4 = 25% ✗ Over-answered instead: 2/4 = 50% Content fully correct: 1/4 = 25%

VERDICT: Worst performance. System barely understands partial evidence should trigger partial answers, not full answers.

INSUFFICIENT EVIDENCE (9 questions, no required units in top-5): Abstention recall: 4/9 = 44% ✗ Over-answered: 5/9 = 56% Content fully correct: 4/9 = 44%

VERDICT: Should abstain, but answers anyway 56% of the time. When it does abstain correctly, the answer is usually still correct.

RESULTS BY QUESTION TYPE

Type A - Single Chunk (6 questions): Calibration: 5/6 = 83% Content correct: 5/6 = 83% Citations correct: 5/6 = 83% VERDICT: Best performance. Simple questions work well. ✓

Type B - Multi-Chunk (5 questions): Calibration: 4/5 = 80% Content correct: 3/5 = 60% Citations correct: 1/5 = 20% ✗ Failure: Synthesis is OK, but citations break. Multiple sources confuse it.

Type C - Partially Answerable (5 questions): Calibration: 1/5 = 20% ✗ (HARDEST) Content correct: 1/5 = 20% Citations correct: 1/5 = 20% VERDICT: Worst type. System fails completely on partial evidence.

Type D - Unanswerable (5 questions): Calibration: 4/5 = 80% Abstention recall: 4/5 = 80% Content correct: 4/5 = 80% VERDICT: Good at recognizing when to abstain (no evidence).

Type E - Dependent-Source (6 questions): Calibration: 5/6 = 83% Content correct: 5/6 = 83% Citations correct: 1/6 = 17% ✗✗ (CITATION DISASTER) Manual citation error: 7/9 = 78% VERDICT: Can answer correctly but citations are severely broken. Clear evidence: system confuses sources when multiple are involved.

CONTINUITY & REPRODUCIBILITY (Q01-Q06)

Questions 1-6 were run in BOTH Experiment B (3B model) and Experiment C. Under greedy decoding, answers should be BYTE-IDENTICAL. They are.

Verification: Verifier: citations 31/32 = 97% ✓ Verifier: misattributed 6/31 = 19% ✓ Manual: misattributed 7/31 = 23% ✓ S4 calibration 6/6 = 100% ✓ S3b justification mean 1.00 ✓

CONCLUSION: Run is reproducible and deterministic. ✓

SENSITIVITY ANALYSIS

Three questions had gold label errors found AFTER the run:

Q10: "Almost onefourth" WAS in top-5, not missed Primary gold: insufficient → Corrected: sufficient Impact on S4: ✗ → ✓ Impact on S3b: 0 → 1

Q18: "Austria (0.029%)" WAS in top-5, wrong source Primary gold: insufficient → Corrected: partial Impact on S4: stays **X** (FULL ≠ PARTIAL+GAP) Impact on S3b: 0 → 0.5

Q22: "8.5 million" WAS in top-5, retrieved correctly Primary gold: insufficient → Corrected: sufficient Impact on S4: **X** → ✓ Impact on S3b: 0 → 1

EFFECT ON POOLED RESULTS (if corrected): S1 sufficient: 14/27 = 52% → 16/27 = 59% S4 calibration: 19/27 = 70% → 21/27 = 78% (+8%) Abstention recall: 4/9 = 44% → 4/6 = 67% S3b mean: 0.70 → 0.82 (+0.12) Taxonomy correct: 33/55 = 60% → 36/55 = 65% Citation metrics: UNCHANGED

REPORTING: Primary results use FROZEN gold as required by the rubric. Sensitivity view is SEPARATE and documented independently.

OUTLIER ANALYSIS (Secondary Results)

Pre-registered outlier rule: >8 claim lines OR ≥400 tokens Q03: 19 lines, 400 tokens (LENGTH LIMIT HIT) Q17: 11 lines, ~350 tokens

Results EXCLUDING outliers (n=25): Metric All 27 Excl. Q03,Q17 Change S2-gen: 91% 91% – S3b mean: 0.70 0.72 +0.02 S4 calibration: 70% 72% +2% S5 over-reach %: 10% 14% +4% Citation incorrect: 24% 30% +6% Taxonomy correct: 60% 60% –

FINDING: Citation error rate goes UP when excluding outliers (24% → 30%) because Q03 had many correct citations. Primary results use all 27.

MAJOR FINDINGS

1. **STRONG EVIDENCE UTILIZATION ✓** When required facts are in top-5, EcoRAG retrieves and uses them 91% of the time. Answers are correct 100% of the time (14/14 sufficient questions).
2. **CRITICAL CALIBRATION FAILURE **XX**** When evidence is absent or incomplete, the model should abstain. It does so **ONLY** 44% of the time (4/9). The other 5 times it answers anyway, usually by substituting related entities or generalizing beyond scope. **THIS IS THE PRIMARY FAILURE MODE.**
3. **CITATION CORRECTNESS IS BROKEN **XX**** 24% of cited claims (20/85) incorrectly point to sources. This is concentrated in dependent-source questions (78% error rate). The system confuses which document supports which claim in multi-source tasks.
4. **RETRIEVAL IS SECONDARY** 18% of required units aren't retrieved (manageable). The bigger problem: even when nothing is retrieved, the model answers.
5. **GENERATION WORKS FINE** Only 5% of failures are due to retrieved units not being used. When evidence is available, the model reads and uses it correctly.
6. **PARTIAL EVIDENCE HANDLING IS WORST** Only 1/4 partial-evidence questions handled correctly. System treats partial evidence like no evidence (over-answers).
7. **DEPENDENT-SOURCE QUESTIONS ARE CITATION DISASTERS** Single-chunk questions: 0% citation error ✓ Dependent-source questions: 78% citation error **X** Clear pattern: multi-source reasoning breaks citation binding.

RESEARCH INTEGRITY

DESIGN: ✓ Frozen before run: questions, gold labels, rubric, retrieval snapshot ✓ No changes during execution: run aborted on snapshot mismatches ✓ Manual review post-hoc: decisions documented separately ✓ A1/A2 decisions applied at review stage, not altering frozen artifacts ✓ Primary (frozen gold) vs Sensitivity (corrected gold) kept separate

REPRODUCIBILITY: ✓ Q01-Q06 byte-identical to Experiment B ✓ Verifier unchanged from Experiments A/B ✓ Greedy decoding (deterministic)

LIMITATIONS: **X** Single reviewer (no inter-rater reliability) **X** Small sample (27 questions, 22 receive full answers) **X** Three post-hoc gold corrections (suggests latent rubric errors) **X** One model configuration (no ablations) **X** One LLM architecture (generalization unknown)

RECOMMENDATIONS

FOR EXPERIMENT D (Improvements):

1. **CALIBRATION-AWARE PROMPT** • Add confidence threshold: "If you don't find the answer in the given passages, say I don't know." • Or use entropy-based abstention: "If confidence < X, abstain."
2. **CITATION BINDING LAYER** • Track source→claim mappings explicitly • Validate each claim against its cited passage before output • Penalize misattributions
3. **MULTI-SOURCE REASONING** • Special handling for dependent-source questions • Explicit source tracking across documents • Contrastive learning on correct vs incorrect bindings
4. **IMPROVED GOLD LABELS** • Second manual reviewer to validate the 85 citation decisions • Continuous (Likert) sufficiency labels instead of binary • Tighter rubric for "unsupported" classification
5. **LARGER EVALUATION SET** • Expand beyond 27 questions • Stratified sampling across document types • Multiple independent reviewers

RESOURCES USED

Documents (9 PDFs, 1,242 chunks): • GMSR 2025 (~500 chunks) • Mapping Green and Digital Energy Jobs (~200 chunks) • Modernising Grids in the Age of Electricity (~180 chunks) • Managing Seasonal Variability of Electricity (~130 chunks) • Critical Minerals Review of Norway 2026 (~80 chunks) • Reducing the Cost of Capital (~60 chunks) • Definitions for Low-Emissions Steel/Cement (~70 chunks) • Manufacturing and Trade Model Documentation (~90 chunks) • UN SG Call to Action on Methane by 2030 (~60 chunks)

System Configuration: • Embeddings: BGE bge-small-en-v1.5 (384 dimensions) • Vector DB: Chroma (1,242 chunks, cosine similarity) • Chunking: v4.4 (512-token max, structural) • LLM: Qwen2.5-3B-Instruct (CPU, bf16) • Prompt: v3 (claim-per-line, max 8 lines) • Retrieval:

Top-5 content/table chunks • Decoding: Greedy (deterministic)

Report generated: 2026-10-04 Classification: Research - Experimental Results Status: Final (frozen gold; sensitivity analysis separate)